Towards building an Indic voice-first conversational bot

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Features of a spoken language system

- It must preserve ***textual intelligence***. However, due to current limitations of voice-based interactions we are only limited to daily scenarios. Performance of spoken dialogue models differ from those of text language models in case solving complex mathematical equations.[Llama]
- An intelligent spoken language system should be able to understand the ***timbre, pitch*** and ***style*** of the conversation and ideally generate response similarly in a zero-shot manner.[Moshi]
- The system should be able to comprehend different ***acoustic information*** and be able to reflect it in its generation. The ability to comprehend and generate acoustic information is a key characteristic of a spoken language system.
- ***Multilinguality*** - It should be able to generate responses in different language and accents.
- For a spoken language system a full-duplex interaction is common. The system should be able to listen and speak simultaneously and respond dynamically to the flow of the conversation.
- The system must be capable of processing and generating responses simultaneous with ***low-latency***.

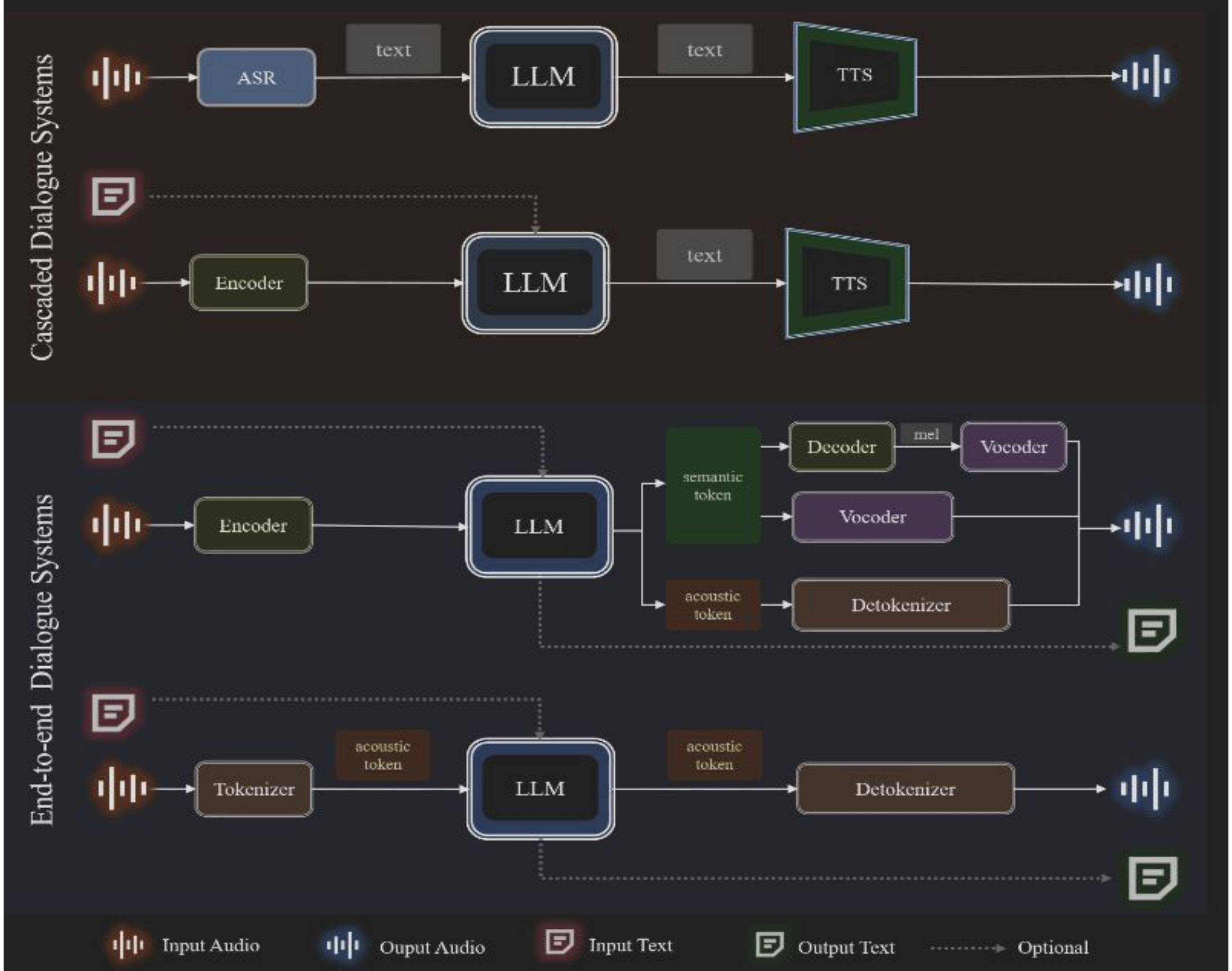


*Image from [WavChat: A Survey of Spoken Dialogue Models]

Types of spoken language system

- ***Cascaded*** - Consists of an ASR module + text-based LLM module + TTS module. In this version, speech is used as only an input-output interface with the system's functionality limited to the text-based LLM. HuggingFace's speech-to-speech has an additional layer of VAD module to distinguish between speech and silent segments and from different speakers. Recently, cascaded systems are integrating speech representations with text as a specialised form of text representation by extending the vocabulary.[Salmonn, Qwen-Audio] The main feature of a cascaded spoken language system is that it only generates text representation at the output. There is a separate TTS module to create the output speech from the text.
- ***End-to-end*** - LLMs that can understand and generate speech representations are classified as end-to-end. Since speech representations are sparser than text, end-to-end systems rely on aligning speech and text modalities whilst preserving textual capabilities of the LLM.[dGSLM]

Representation of speech tokens

- ***Semantic*** - represents linguistic information which allows speech to be mapped to embedding layers of LLMs, facilitating modality alignment and leveraging the strengths of LLM. They can be compressed to high rates. However, these representations do not capture naturalness and expressiveness of speech properly.
- ***Acoustic*** - represents the more nuanced emotional information(timbre, pitch, style) which are important in creating expressive speech. Acoustic representations are less conducive to LLM understanding. They are designed to keep reconstruction in mind which makes high compression and high quality reconstruction challenging to achieve simultaneously.

Modality alignment

- ***Text output only*** - speech input is mapped into the latent text space using a speech encoder and an adaptor. This method of embedding aligning equips the LLM with understanding and processing speech inputs. However, the output is still text-based which requires a TTS module to generate speech.
- ***Chain-of-modality*** - This method tokenizes speech into discrete tokens and extends the LLM's vocabulary to handle both speech input and output. It generates response text autoregressively before producing the corresponding speech. This technique allows the text LLM's output to guide speech generation, thereby enhancing the quality of the response content. However, it is not suitable for live interactions, as the model must complete the entire text response before beginning speech generation, leading to increased response latency.
- ***Interleaving text and speech tokens*** - Spirit-LM interleaves speech tokens with text tokens leveraging on the temporal alignment between its speech input and the transcribed text to merge modalities.

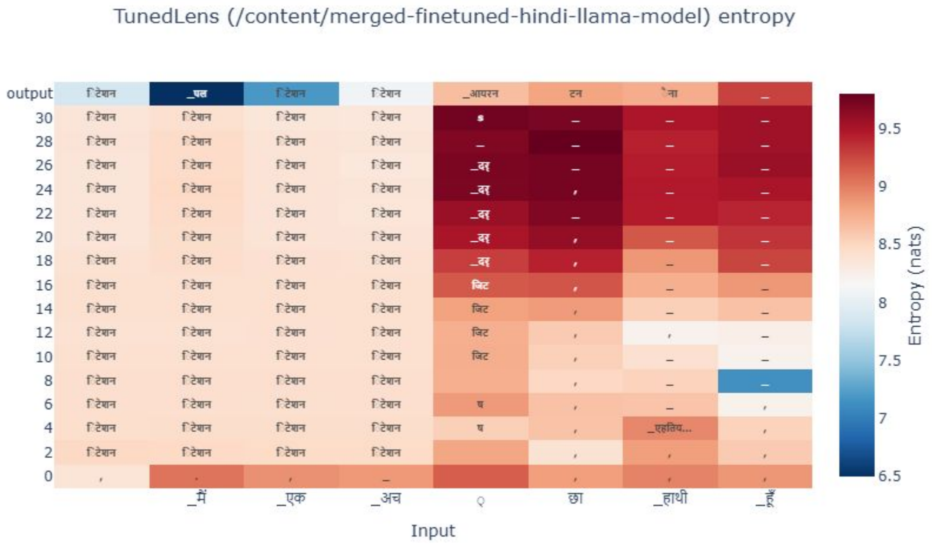
Proposed design

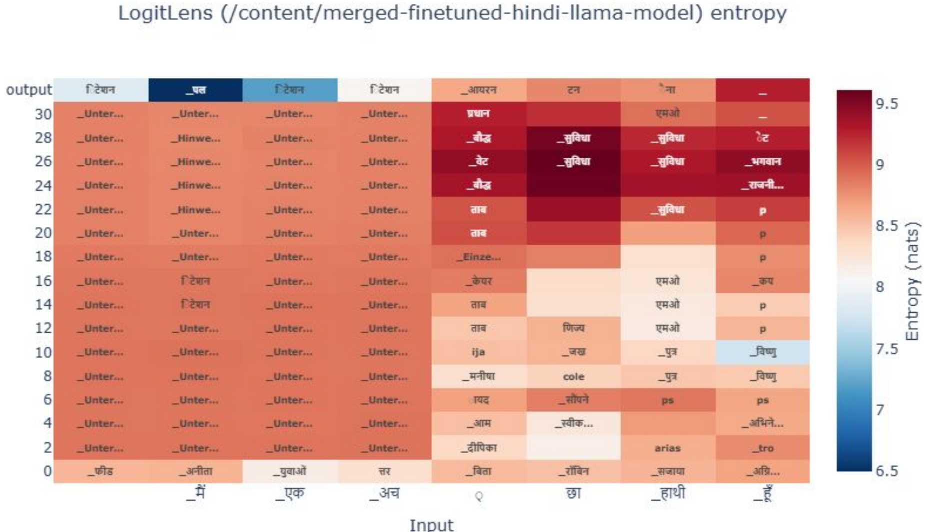
- Start with a multilingual LLM because we intent to preserve the textual capabilities of the LLM and aide in spoken language generation.
- We will design an end-to-end system with parallel generation of speech and text to reduce latency and maintain a conversational environment.
- Map speech tokens with text tokens. LLMs have shown multilingual capabilities even when not trained in those languages due romanised representation.[Saji et.al]
- Proposition is to segment speech tokens syllable-wise and map them into the inner hidden representation of text tokens. Benefits - We can universally ensure equivalent mutual information in speech-text across languages.

Experiment

- I finetuned a Llama-2 model with 1043 Hindi paragraphs sampled from the Rahular Varta dataset. I followed the finetuning instructions mentioned in Llama-cookbook.[Code]
- I trained a tuned lens on the merged llama model, tokeniser, and data for visualising the inner layer representation of "मैं एक अच्छा हाथी हूँ".[Code]

  All the code and results are packaged in this repository.





Conclusion

- A fine-tuned Llama model doesn’t represent characters in the romanised form. The tuned lens visualisation showed the inner representations of the characters as it was loaded with the trained lens. The logit lens could only load a pre-trained lens hence, the inner representations show unrecognised characters.
- Hence the initial proposition to ensure speech-text alignment across languages won't be possible. We will have to validate it for every single language the model is being trained on. There could be a possibility of slight variation in speech-text mutual information representation which may lead to minor discrepancies in recognition and understanding part of a voice-first conversational bot.
- However, since we will be working mostly with Llama models, we can always try interleaving of speech & text tokens (at the word-level) as demonstrated in Spirit-LM for 22 indian languages and english to merge modalities.

Proposed methodology

- We finetune a Llama-3+ model on a Instruction-tuned dataset (child-safe + educational content) covering 22 indian languages + english. We will be leveraging on its textual capabilities and the way the model learns the data. For now we can use IndicIFEval.
- Since there are transcripts present for the speech dataset, we use interleaving of speech and text tokens (like Spirit-LM) to merge speech-text modality and achieve the required alignment. Spiri-lm has noted that interleaving speech and text tokens generalises well on what is learned during Llama2 pretraining. Interleaving preserves the right-to-left natural causality of the data within each modality and also across modalities, allowing the model to learn aligned representation between speech and text units.
- In case, the above method does yield expected results, we can always start with training RVQ layers with Hubert representations of the input speech and achieve the required speech-text alignment as proposed in SpeechTokeniser.

Translation ability in spoken language model

- We can follow the methods in Futami et.al to predict the target speech tokens (semantic token) and minimise the below loss function

  $L = -\log p(S^{tgt}|T^{tgt}, T^{src}, S^{src})\ p(T^{tgt}|T^{src}, S^{src})\ p(T^{src}|S^{src})$

  where S^{src} = semantic token of the source language

  S^{tgt} = semantic token of the target language

  T^{src} = text token of the source language

  T^{tgt} = text token of the target language

- In the above paper, the authors have used interleaving of speech-text units gradually reducing the ratio of text units with each iteration for a seamless transfer of modality from a text LM to a speech LM.

Expressive-ness

We can supplement speech tokens with pitch and style tokens and finetune the LLM to produce more expressive speech.

Spirit-LM had an input sequence like

[SPEECH][St10][Pi0][Hu28][Hu22][Pi14][Hu15][Pi32][Hu78][Hu234][Hu468]

Thank you😊 ✌️